AIC03

Single-Stage Amplifiers and Source Followers

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Complete both diode-load CS circuits, the current-source-load CS and N/P source followers from *AIC Lab.pdf*, PDF pages 31–40. Check geometry, bias, output resistance, gain and bandwidth for each case. The model library is supplied separately by the user. The 1.0 V supply is a teaching adaptation, not a certification of the model voltage rating. Numerical plots for the handout's original 3.3 V process are not used directly as pass thresholds. The circuits are rebuilt from the handout topology; the earlier code is used only to review dimensions and omissions.

2 Specification Mapping

S denotes actual simulation and D denotes derivation/discussion. The old-process gain tables are background comparisons, not guarantees for the supplied model.

Requirement and PDF pages	Deck / check	Status
PMOS diode-load CS, W=100/200/300 (31–32)	D01; DC/AC/Rout	S+D
NMOS diode-load CS, W=150/200/300 (33–34)	D02; DC/AC/Rout	S+D
Diode-load gain, gain-bandwidth and tradeoffs (34)	D01–02; derivatives	S+D
Current-load CS, equal-W/L scaling and mirror accuracy (35–37)	D03/C0,C5,C6	S+D
NMOS/PMOS source followers (38–39)	D04; gain/BW	S+D
CS W=10/18/26 and Iref=100/166/333 (39–40)	D03/C0–C4	S+D
Low-current adaptation and time-domain check	D03/C7–C9, D05	S; adapted

3 Circuit and Design Derivation

$$A_{v,P} \simeq -\frac{g_{m1}}{g_{ds1}+g_{m2}+g_{ds2}}, \quad A_{v,N} \simeq -\frac{g_{m1}}{g_{ds1}+g_{m2}+g_{mb2}+g_{ds2}}.$$

$$A_{v,CS} \simeq -\frac{g_{mn}}{g_{dsn}+g_{dsp}}, \qquad A_{v,SF} \simeq \frac{g_m}{g_m+g_{mb}+g_{ds,F}+g_{ds,load}}.$$

The derivative estimate uses each case's actual `.op` values, not an assumed mobility. R_o is obtained from the small-signal `.tf` output resistance, with capacitors open at DC. $GB = |A_v(1\,\text{Hz})|f_{3dB}$ is

stored separately from the actual unity crossing; the two are not forced to agree in a multipole circuit.

D01/D02 retain the handout $W/L = 2/1\,\mu$m load and $C_L = 10$ pF, sweep the specified input-device widths, and use $L = 1\,\mu$m. The page 33 statement tying an NMOS body to VDD is inconsistent with a conventional P-substrate implementation; the body is grounded here and g_{mb} is retained. These two diode-load topologies were absent from the earlier code and are now included.

D03/C0–C4 use $(W_n, W_p, L, I_{ref}) = (10, 30, 1, 100)$, $(18, 30, 1, 100)$, $(26, 30, 1, 100)$, $(10, 30, 1, 166)$ and $(10, 30, 1, 333)$, with dimensions in μm and current in μA. C5/C6 multiply every W and L of C0 by 3/5, preserving W/L. C7–C9 are explicit 10 μA adaptations with $W_n = 10/18/26\,\mu$m. Each CS input bias is calibrated from its DC transfer and local derivative to approximately 0.5 V output; no servo amplifier is added to the device under test.

D04 uses $W/L = 30/1\,\mu$m and $C_L = 10$ pF for both followers, with NMOS/PMOS gate biases of 0.9/0.1 V. C0–C3 use 100/10/166/333 μA. Consistent circuits across analyses replace the earlier mismatched DC/AC/transient dimensions and beyond-rail input signals.

4 Simulation Setup

Runs use PrimeSim HSPICE S-2021.09 win64, a baseline temperature of 27 °C, `runlvl=5`, `accurate`, and `.op/.print/.measure` outputs. Every `.alter` resets all baseline parameters and temperature to prevent state from carrying into the next case. Actual runs completed without an error or failed measurement. Suppressed model warnings remain, so a warning-free run is not claimed. Public netlists contain no model paths. This comment marks the model setup required locally before execution; the model and its configuration remain local:

```
* MODEL_SETUP_REQUIRED
```

CSV files and numerical summaries are in the lab's `results/` directory. Every figure is redrawn from actual printed data, not a waveform-viewer screenshot. `manifest.json` supplies specification mapping and case definitions.Five decks contain 23 cases. CS DC sweeps use 0–1 V in 1 mV steps; AC uses 1 Hz–10 GHz, 60 points/decade and a 1 V input. The follower DC sweep changes only the NMOS input; PMOS stays at 0.1 V. Its N/P AC circuits are independent, avoiding normalization against oppositely phased inputs. D05 runs a 1 kHz signal for 5 ms and measures peak-to-peak output over the last period. CS amplitudes are 0.1/1/3 mV; follower amplitude is 10 mV.

5 Results and Specification Checks

Case	V_{in} (V)	$\lvert A_v \rvert$	g_m estimate	R_o (kΩ)	f_{3dB} (MHz)	Sat.
D01/C0	0.21622	2.35	2.35	43.07	0.366	Y
D01/C1	0.18785	2.41	2.41	43.05	0.363	Y
D01/C2	0.17159	2.44	2.45	43.04	0.360	Y
D02/C0	0.19697	1.11	1.11	21.35	0.736	Y
D02/C1	0.18532	1.13	1.13	21.35	0.733	Y
D02/C2	0.16909	1.14	1.14	21.35	0.727	Y
D03/C0	0.55243	124.30	124.34	126.10	0.125	Y
D03/C1	0.49981	156.65	156.69	125.30	0.126	Y
D03/C2	0.47225	174.09	174.13	123.60	0.128	Y
D03/C3	0.60983	69.37	69.39	54.73	0.289	Y
D03/C4	0.71517	11.10	11.10	6.68	2.363	N
D03/C5	0.55481	179.14	179.92	181.20	0.086	Y
D03/C6	0.55539	201.82	204.63	204.10	0.075	Y
D03/C7	0.39126	227.10	227.14	1208.00	0.013	Y
D03/C8	0.36092	233.00	233.05	1159.00	0.014	Y
D03/C9	0.34283	234.85	234.92	1133.00	0.014	Y

Sat.=Y only means that every device at this nominal operating point satisfies $|V_{DS}| \geq |V_{DSAT}|$. It does not imply strong inversion or valid operation across the entire output swing. Subthreshold labels remain in `operating_points.csv`. The diode-load CS devices often enter weak inversion under these biases, so the strong-inversion $\sqrt{W}$ gain scaling must not be imposed.

I_{ref} (μA)	A_n	A_p	f_n (MHz)	f_p (MHz)
100	0.8260	0.6855	27.032	12.355
10	0.8337	0.8503	3.992	2.856
166	0.8217	0.5344	38.088	17.654
333	0.8079	0.4060	57.736	25.284

Input amplitude (mV)	CS output p-p (V)	Effective $\lvert A_v \rvert$	Follower p-p (V)
0.1	0.04521	226.04	0.01667
1	0.42794	213.97	0.01667
3	0.74588	124.31	0.01667

D03 case	Mirror error (%)	GB (MHz)
0	-0.492	15.594
1	-0.492	19.764
2	-0.492	22.256
3	-0.968	20.051
4	-3.926	26.230
5	-0.337	15.450
6	-0.292	15.224
7	0.392	2.976
8	0.392	3.180
9	0.392	3.278

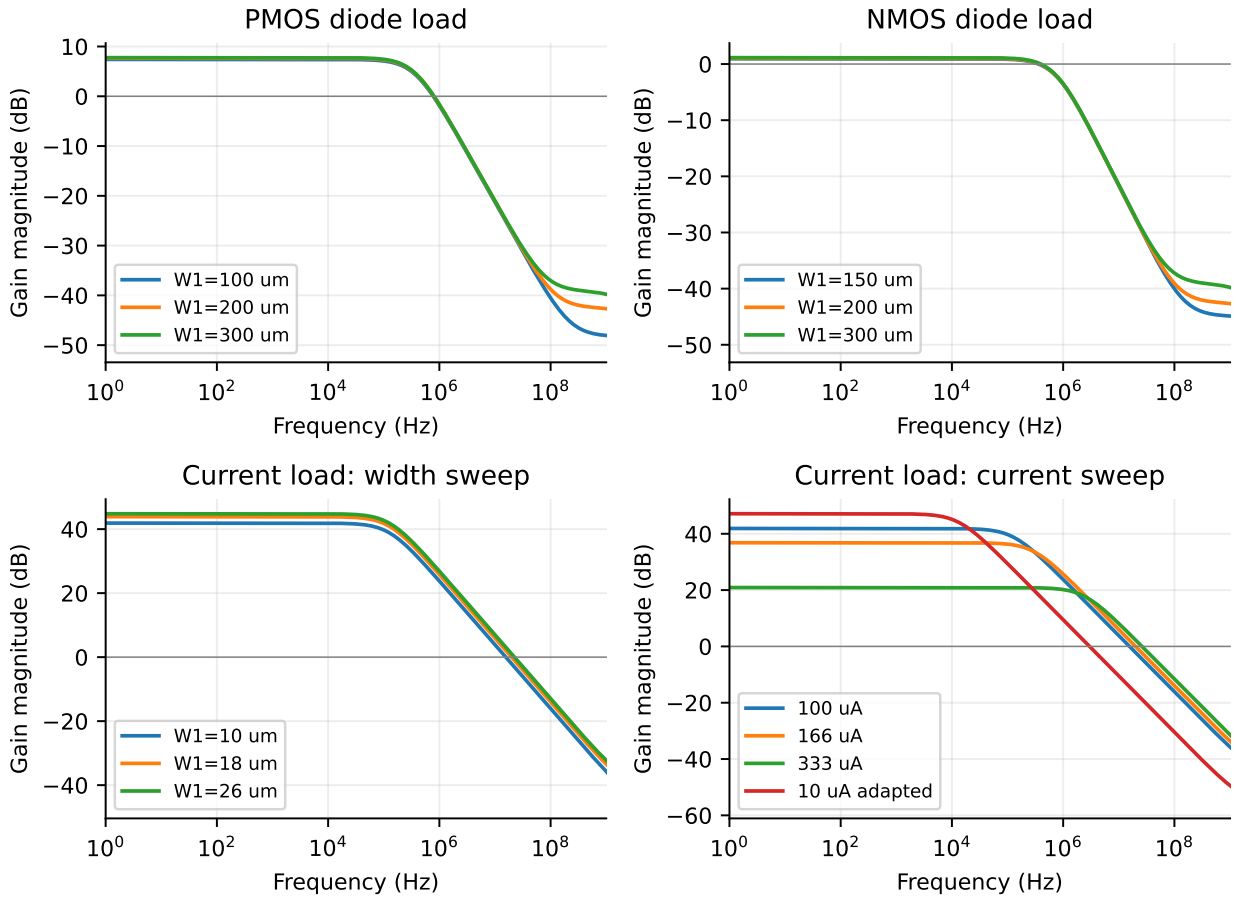


Figure 1: Actual Bode curves for both diode loads and current-source loads. Each geometry/current comparison retains its own operating point.

D03 mirror error is $(|I_{MP}|/|I_{MREF}|-1)\times 100\%$. The 333 μA case is retained, but reduced reference headroom and device operating regions limit its gain; width alone does not explain the result. D05 with 0.1 mV input approaches the small-signal response. Larger amplitudes reduce effective gain because of DC-transfer curvature.

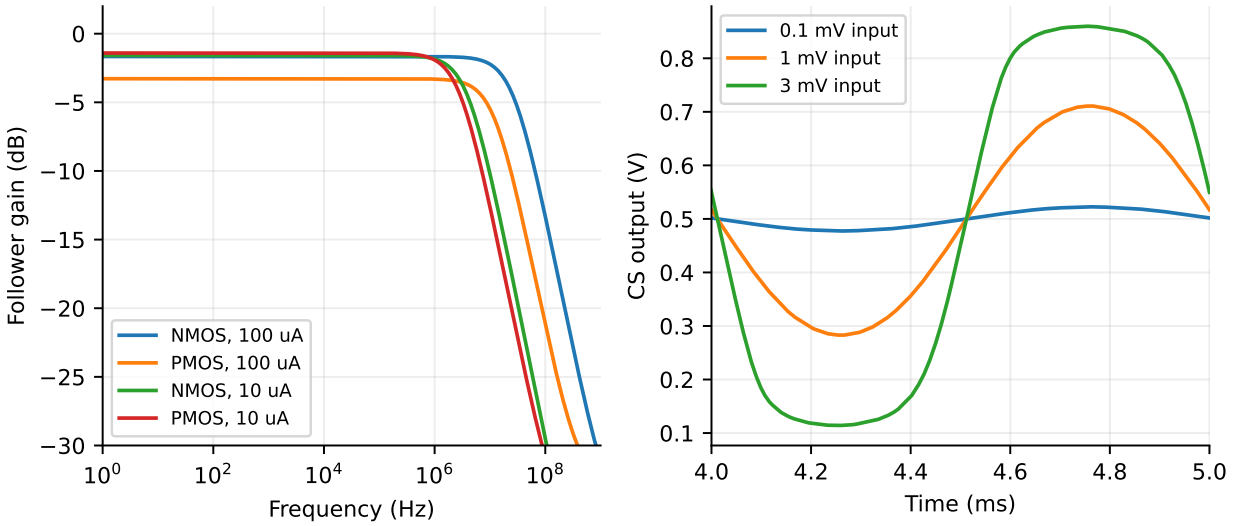


Figure 2: N/P follower frequency responses and the CS amplitude comparison. The right panel shows actual output during the final 1 kHz period.

6 Discussion

For equal load geometry, mobility, bias and the absence of the same NMOS source-body term can give the PMOS diode load a smaller load transconductance and greater CS gain. This is conditional, not universal across processes and biases. Diode loads are simple to bias, use few devices and give low output resistance, at the expense of gain and voltage headroom. Increasing input g_m or reducing effective load admittance can raise gain; more current or less load capacitance can improve speed while also changing power, output resistance and bias.

A current-source load removes the load g_m term, increasing gain but making the output bias more sensitive to current matching. At fixed current, larger W can increase strong-inversion g_m while also adding capacitance. Increasing current changes both g_m and g_{ds} and need not increase gain. Longer L at equal W/L can reduce channel-length modulation and improve mirror accuracy, but costs area and parasitic capacitance.

Body effect and finite load conductance reduce source-follower gain. NMOS is closer to unity at 100 μA, but at 10 μA this PMOS result is approximately 0.8503, higher than NMOS at approximately 0.8337. The N-type advantage in the old plot is not a universal rule.

7 Conclusion

The specified topologies and width, current and equal-W/L scale variations have been simulated, including the previously missing diode loads. Gain, bandwidth and nonlinear results apply to their declared biases. Old-process numerical agreement and full-PVT success are not claimed.

8 MOS transistor schematic

These editable vector schematics connect equal node names within a circuit. Primitive-cell names are local pins; the hierarchy defines their connections. B denotes the body terminal; 0 is global ground. Each panel identifies its netlist and scope; measurement sources are connected as described. Complete current-source-loaded common-source stage from `03_current_source_load.sp`.

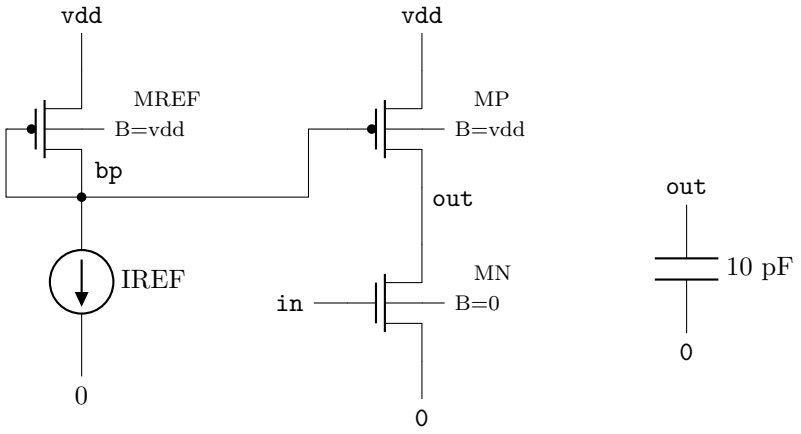


Figure 3: PMOS current mirror biases the NMOS common-source amplifier.

The diode-connected MREF sets bp; MP shares that gate voltage. PMOS sources and bodies connect to vdd=1 V; MN source and body connect to 0. Baseline MN $W = 10\,\mu$m, MREF/MP $W = 30\,\mu$m, all $L = 1\,\mu$m. IREF flows from bp to 0 with baseline 100 μA. VIN drives in relative to 0 at the deck's bias parameter vb with unit AC amplitude. The load capacitor is connected from out to 0. Geometry/current sweeps preserve this topology. Diode-load and source-follower comparisons are separate topologies in the other source decks; they are not represented by this figure.